

# Distribution Assignment

**1. Simulate 30 rolls with =RANDBETWEEN(1,6). What is the probability of rolling a 3 exactly 5 times? (Hint: Use BINOM.DIST)**

**Answer:**

In Excel, enter:

`=RANDBETWEEN(1,6)`

and generate 30 random rolls.

To calculate the theoretical probability of rolling a 3 exactly 5 times, use the BINOM.DIST function:

`=BINOM.DIST(5,30,1/6,FALSE)`

Here:

- Number of trials (n) = 30
- Number of successes (x) = 5
- Probability of rolling a 3 (p) = 1/6
- Cumulative = FALSE because we need exactly 5 times

The result is approximately:

**0.1921**

Therefore, the probability is approximately:

**19.21%**

**Answer: 0.1921 or approximately 19.21%.**

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**2. Generate 100 values in Excel using the continuous uniform distribution RAND() and plot a histogram. Describe the shape of the distribution.**

**Answer:**

In Excel, enter the following formula in a cell:

`=RAND()`

Copy the formula down until you have **100 values**.

The RAND() function generates random values between 0 and 1, following a continuous uniform distribution.

To create the histogram:

1. Select the 100 generated values.
2. Go to **Insert**.
3. Select **Histogram** from the Charts section.
4. Create the histogram.

#### **Description of the distribution:**

The histogram should have an approximately **uniform shape**, because values between 0 and 1 have approximately equal chances of occurring. The bars should be relatively similar in height, although there may be some small differences because the values are randomly generated.

**Answer: The distribution is approximately uniform, with values spread fairly evenly between 0 and 1.**

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**3. A dataset has a mean of 50 and a standard deviation of 5. What percentage of values lie between 45 and 55 if the data follows a normal distribution?**

**Answer:**

Given:

Mean ( $\mu$ ) = 50

Standard deviation ( $\sigma$ ) = 5

The lower value is 45:

$$z = (45 - 50) / 5$$

$$= -1$$

The upper value is 55:

$$z = (55 - 50) / 5$$

$$= 1$$

Therefore, the range from 45 to 55 represents **one standard deviation below and one standard deviation above the mean**.

According to the empirical rule for a normal distribution, approximately **68.27%** of the values lie within  $\pm 1$  standard deviation of the mean.

**Answer: Approximately 68.27% of the values lie between 45 and 55.**

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**4. What is the concept of standardization (z-score), and why is it important in data analysis? Explain the formula and how standardization transforms a dataset.**

**Answer:**

Standardization is the process of transforming data so that it has a **mean of 0 and a standard deviation of 1**. The standardized value is called a **z-score**.

The formula for the z-score is:

$$z = (x - \mu) / \sigma$$

Where:

- $x$  = individual data value
- $\mu$  = mean of the dataset
- $\sigma$  = standard deviation

A z-score tells us how many standard deviations a particular value is above or below the mean.

For example, if a value has a z-score of +2, it is two standard deviations above the mean. If it has a z-score of -1, it is one standard deviation below the mean.

**Importance of standardization:**

- It puts variables with different scales onto a common scale.
- It makes comparison between different variables easier.
- It is useful in statistical analysis and machine learning.

- It prevents variables with larger numerical values from having an unnecessary influence on analysis.

### **How standardization transforms a dataset:**

After standardization, the transformed dataset has a mean approximately equal to **0** and a standard deviation approximately equal to **1**.

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## **5. What is Kurtosis and their type?**

### **Answer:**

**Kurtosis** is a statistical measure that describes the shape of a probability distribution, particularly the **heaviness of its tails and the concentration of values around the mean**.

There are three main types of kurtosis:

### **1. Mesokurtic**

A mesokurtic distribution has a shape similar to the normal distribution. It has moderate tails and moderate concentration around the mean.

**Example:** Normal distribution.

### **2. Leptokurtic**

A leptokurtic distribution has heavier tails and a greater concentration of values around the mean compared with a normal distribution. It is more likely to contain extreme values or outliers.

### **3. Platykurtic**

A platykurtic distribution has lighter tails and a flatter shape compared with a normal distribution. Extreme values are less common.

**Answer: The three main types of kurtosis are Mesokurtic, Leptokurtic, and Platykurtic.**

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## **6. Explain why the uniform distribution is a good model for the outcome of rolling a fair die.**

**Answer:**

A uniform distribution is a good model for rolling a **fair die** because each possible outcome has an equal probability of occurring.

For a standard six-sided die, the possible outcomes are:

**{1, 2, 3, 4, 5, 6}**

The probability of each outcome is:

$$P(1) = 1/6$$

$$P(2) = 1/6$$

$$P(3) = 1/6$$

$$P(4) = 1/6$$

$$P(5) = 1/6$$

$$P(6) = 1/6$$

Since all six outcomes have the same probability, the distribution is uniform.

Therefore, a fair die follows a **discrete uniform distribution**.

**Answer: The uniform distribution is suitable because every outcome of a fair die has an equal probability of 1/6.**

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**7. Use Excel to compute the probability of getting at least 8 successes in 15 trials with success probability 0.5**

**Answer:**

Given:

Number of trials (n) = 15

Success probability (p) = 0.5

Required successes = At least 8

“At least 8” means:

**8, 9, 10, 11, 12, 13, 14, or 15 successes.**

In Excel, use:

**=1-BINOM.DIST(7,15,0.5,TRUE)**

This calculates the probability of getting more than 7 successes.

The result is:

**0.5**

Therefore:

**Probability = 0.5**

or

**Probability = 50%**

**Answer: 0.5 or 50%.**

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**8. How does log transformation help in stabilizing variance and making data more normally distributed?**

**Answer:**

Log transformation is a mathematical transformation in which the logarithm of each data value is taken.

The transformation can be written as:

$$y = \log(x)$$

It is particularly useful for data that is **right-skewed** or has values that vary greatly in size.

Log transformation helps by:

- Reducing the effect of very large values.
- Compressing the scale of the data.
- Reducing right skewness.
- Making the distribution more symmetric.
- Helping to stabilize the variance when the variability increases with the mean.
- Making some statistical methods that assume approximately normal data more appropriate.

For example, income, population size, and sales data can sometimes have highly skewed distributions. Applying a log transformation can make these distributions more balanced and easier to analyze.

**Answer: Log transformation compresses large values, reduces skewness, and can stabilize variance, making the data more suitable for statistical analysis and sometimes closer to a normal distribution.**